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***B.Tech. Degree I Semester Regular/Supplementary and  
II Semester Supplementary Examination March 2022***

**ME/SE 19-200-0103A ENGINEERING GRAPHICS**

*(2019 Scheme)*

Time: 3 Hours

Maximum Marks: 60

**PART A**

$(1 \times 12 = 12)$

- I. Construct a diagonal scale with RF 1/5000 showing meters and long enough to measure up to 1 kilometer. Mark a distance of 658 meters on it. (12)
- OR**
- II. The major axis of an ellipse is 150 mm long and the minor axis is 100 mm long. Find the foci and draw the ellipse by arc of circle or intersecting arc method. Draw a tangent and normal to the ellipse at a point on it. (12)

**PART B**

$(4 \times 15 = 60)$

- III. Top view and front view of a line AB 80 mm long measure 70 mm and 40 mm respectively. End A is 15 mm above HP and 25 mm in front of the VP. Draw the projections of the line and find the true inclinations with HP and VP. (15)
- OR**
- IV. (a) A square plane ABCD of 30 mm side has its surface parallel to H.P. and 20 mm above it. Draw its projections of the plane when two of its sides are (5)
- (i) parallel to V.P.
- (ii) inclined at  $30^\circ$  to V.P.
- (iii) all sides are equally inclined to V.P.
- (b) Draw the projections of a circular thin plate of diameter 10 mm resting on the ground on a point A on the circumference, its surface is inclined at  $45^\circ$  to the H.P. and the plan of the diameter AB is making  $30^\circ$  with VP. (10)
- V. Draw the projections of a cone, base 45 mm diameter and axis 50 mm long, when it is resting on the ground on a point on its base circle with (15)
- (i) the axis making an angle of  $30^\circ$  with the H.P. and its top view making  $45^\circ$  with the V.P.
- (ii) the axis making an angle of  $30^\circ$  with the H.P. and  $45^\circ$  with the V.P.
- OR**
- VI. A pentagonal prism of base side 30 mm and axis length 60 mm is resting on HP on one of its rectangular faces, with its axis perpendicular to VP. It is cut by a plane inclined at  $50^\circ$  to VP and perpendicular to HP and passing through a point 25 mm from rear base of the prism. Draw its top view, sectional front view and true shape of section. (15)
- VII. A hexagonal pyramid of base side 25 mm and height 60 mm rests vertically on HP with one of the base sides perpendicular to VP. It is cut by a section plane inclined at  $45^\circ$  to the HP, perpendicular to the VP and bisecting the axis of the pyramid. Draw the development of the bottom part of the pyramid. (15)

**OR**

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- VIII. A vertical cylinder of 40 mm diameter and 80 mm height is resting on its base. A horizontal cylinder of 30 mm diameter and 70 mm height and with the axis parallel to both the planes, 40 mm above HP and 3 mm in front of the axis of the vertical cylinder penetrates through the vertical cylinder. Draw the curves of intersection. (15)
- IX. A hemisphere of 80 mm diameter is nailed on the top of pentagonal prism with flat surface up. Base edge of the prism is 60 mm and height is 120 mm. The axes of the both solids coincide. Draw the isometric projection. (15)
- OR**
- X. A pentagonal pyramid of height 45 mm and base edge 30 mm is resting on its base with one base edge parallel to and 10 mm behind PP. The station point is 22 mm in front of PP, 38 mm to the left of the axis and 55 mm above GP. Draw the perspective view. (15)

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